Village Economic Growth Intelligence — Technical Documentation

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Overview

The pipeline ranks India's villages on satellite-observed growth from 2020/21 to 2025 and returns one list of the top 100 villages.

Definition of economic growth. A sustained rise in economic *activity* (night-time lights) together with new physical *capital* (persistent new built-up area), measured against each village's own starting point.

Item	Value
Unit	Census 2011 village polygon (SHRUG 2.2), 639,782 eligible villages
Period	Land cover 2020 → 2025; night lights 2021 → 2024–25
Signals	Night-light trend, night-light gain, new built-up share, settlement expansion
Tools	Google Earth Engine (GEE) for raster statistics; Python for preparation and scoring
Output	`data/output/top100_villages.csv`

Data sources

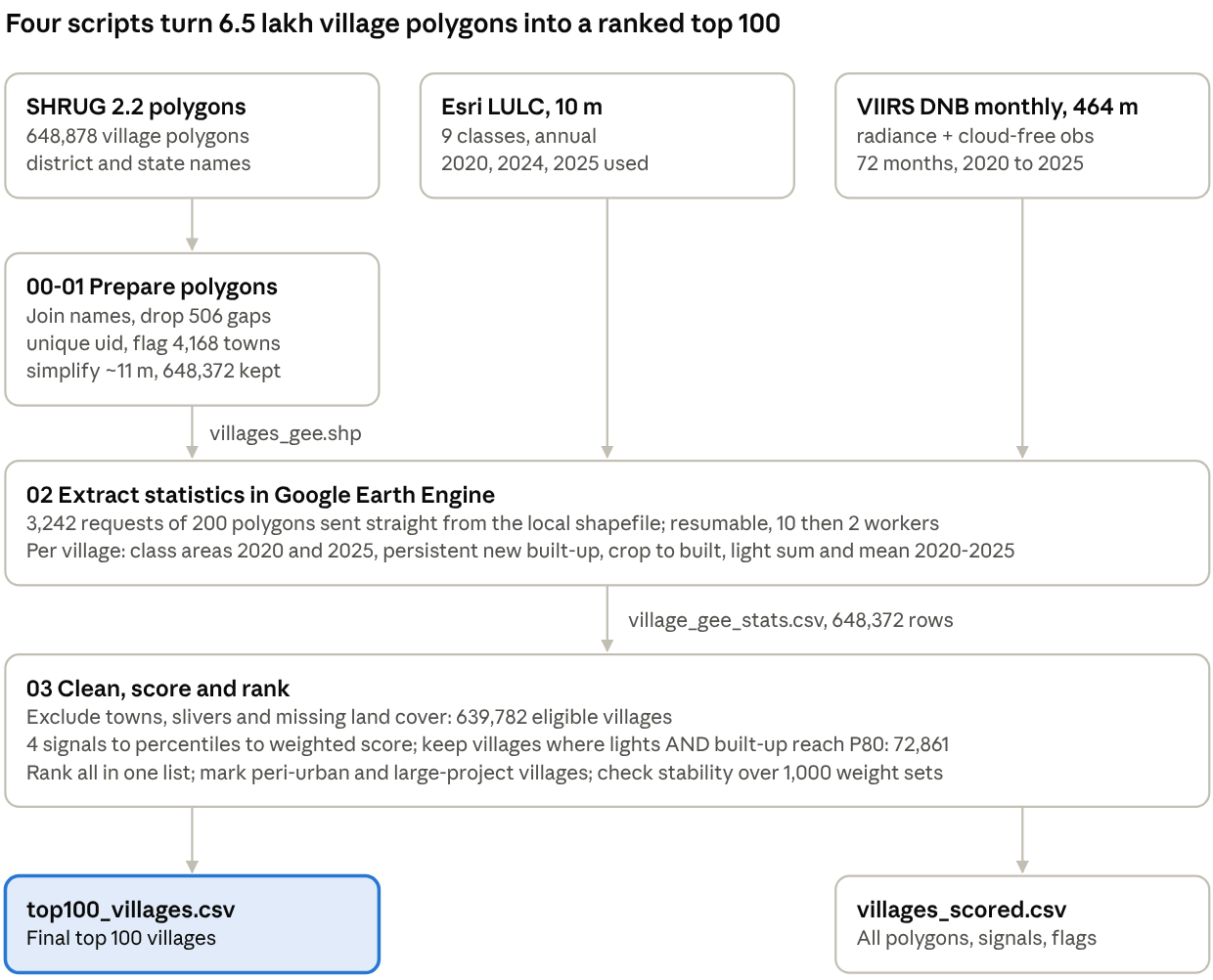
Three open datasets carry the whole analysis; each was chosen because it covers all of India consistently across the study years.

Dataset	What it gives	Resolution	Years used	Why chosen	Access
SHRUG 2.2 "Pakora" open polygons (Development Data Lab)	PC11 village/town polygons (648,878), plus district (641) and state (35) polygons with names	Vector; boundaries carry ~0–1 km error	Census 2011	Only open all-India village boundary set; PC11 codes link to Census and SHRUG socio-economic tables	Direct download, CC BY-NC-SA 4.0
Esri 10 m Annual LULC (Impact Observatory / Esri, Sentinel-2)	9 land-cover classes per year incl. Built Area and Crops	10 m	2020, 2024, 2025	Consistent annual series 2017–2025 from one model; 10 m resolves village-scale construction	GEE asset `projects/sat-io/open-datasets/landcover/ESRI_Global-LULC_10m_TS`
VIIRS DNB monthly composites, VCMSLCFG (Earth Observation Group, Colorado School of Mines)	Average night-time radiance + cloud-free observation count per month	~464 m	2020–2025 (72 months)	Standard proxy for economic activity; monthly series runs to Sep 2026, so 2025 is complete	GEE `NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG`

The official annual VIIRS product (VNL v2.2) stops at 2024. All six annual night-light layers were therefore built from the monthly product with one method, so no year mixes two processing chains.

Pipeline architecture

Four scripts run in sequence; only step 02 calls GEE, and each step writes the file the next one reads.



pipeline · 3 sources, 4 scripts, 2 outputs

```
Kritter/
├── scripts/    00_add_admin_names.py, 01_prepare_gee_upload.py,
02_extract_gee.py, 03_score_villages.py
├── data/
|   ├── raw/        SHRUG downloads
|   ├── interim/    prepared polygons, village_gee_stats.csv
|   └── output/     top100_villages.csv, villages_scored.csv
└── logs/       GEE run logs
```

Step 0–1: Village boundary preparation

Two local scripts turn the raw SHRUG file (648,878 polygons) into 648,372 named, uniquely keyed polygons.

1. **Add names** (`00_add_admin_names.py`): join state and district names from the SHRUG state and district files on Census codes (an attribute join, not a spatial one).
2. **Drop 506 non-villages**: polygons with village code `000000` are unnamed gaps such as forest blocks and uninhabited islands.
3. **Unique ID**: `uid` = state_district_subdistrict_village code, because 155 village codes repeat where a village is split across sub-districts.
4. **Flag towns**: Census town codes start with 8; 4,168 town polygons are flagged and later excluded.
5. **Simplify** boundaries by ~11 m, well inside the source's own ~1 km boundary error.

Output: `data/interim/gee_upload/villages_gee.shp`.

Step 2: GEE extraction

`02_extract_gee.py` sends the local polygons to GEE in 3,242 batches of 200 and returns 33 statistics per village; no asset upload is needed.

Land cover (Esri LULC, 10 m)

- Area in hectares of each of the 9 classes, 2020 and 2025.
- **New built-up** = not Built in 2020 and Built in **both** 2024 and 2025. The two-year rule removes single-year class flicker.
- **Crop → built** = Crops in 2020 that became persistently Built.

Night lights (VIIRS monthly, ~464 m)

- Months with fewer than 2 cloud-free observations are dropped.
- Annual value = median of the remaining months, which also removes short-lived lights such as fires.
- Values below 0.5 nW/cm²/sr are set to 0 (background noise).
- Per village and year, 2020–2025: total light (area-weighted sum) and mean radiance.

The run is resumable: each batch is saved as it finishes and skipped on restart. Output: `data/interim/village_gee_stats.csv` (648,372 rows).

Step 3: Scoring and ranking

`03_score_villages.py` scores 639,782 eligible villages on four signals, keeps the 72,861 where both lights and construction rise strongly, and ranks them in one list.

Exclusions. Towns (4,168), fragments under 10 ha (4,409), and 13 villages with missing Esri land cover (Lakshadweep in 2020, Narcondam Island).

Signals

Signal	Formula	Weight
Night-light trend	Theil-Sen slope of ln(1 + total light), 2021–2025	0.30
Night-light gain	(mean total light 2024–25 − 2021) per km²	0.20
New built-up share	new built-up ha / village area ha	0.25
Settlement expansion	new built-up ha / (2020 built-up ha + 1)	0.25

2021 is the night-light baseline because the 2020 COVID lockdown depressed lights nationwide. Each signal becomes a 0–100 percentile among eligible villages, and the score is their weighted mean:

```
\text{score} = \frac{\sum_k w_k \, p_k}{\sum_k w_k}
```

Agreement filter. A village is ranked only if a night-light signal **and** a built-up signal both reach the 80th percentile, with at least 1 ha of new built-up and a lit end period. This removes single-sensor artefacts.

Attributes for interpretation (not used in the score)

- `peri_urban` = 1 if the village is within 10 km of one of the 500 brightest towns, a stand-in for cities of over 1 lakh people.
- `flag_large_project` = 1 if the change looks like a single project rather than organic growth: ≥ 100 ha new built-up; or ≥ 30% of the village newly built from a small base (< 25 ha, ≥ 15 ha new); or lights up ≥ 15× to a substantial level.
- `flag_crop_loss` = 1 if more than 30% of the village lost cropland to something other than buildings, a sign of crop/rangeland misclassification.

Rank stability. The ranking is repeated with 1,000 random weight sets; `rank_stability` = % of them that keep the village in the top 100.

Results

The top 100 villages span 18 states; 31 sit within 10 km of a major town and 23 are driven by a single large project.

National change, all villages

Measure	2020	2025	Change
Built-up area	19.7 Mha	23.4 Mha	+19%
New built-up (persistent)	–	3.56 Mha	2.39 Mha of it was cropland
Total night light	12.6 M	19.7 M	+57%
Villages with any light	89.1%	96.6%	+7.5 pts

Top 100

- States: Karnataka 16, Uttar Pradesh 13, Odisha 10, West Bengal 9, Rajasthan 8, Madhya Pradesh 7, Haryana, Jammu and Kashmir and Himachal Pradesh 5 each.
- Median village: 228 ha, 36 ha of new built-up, night light 5.3× its 2021 level, 18 km from a major town.
- Scores 97.9–99.9; median rank stability 67%, with 29 villages below 50%.

Rank	Village	District, State	New built-up (ha)	Total light 2021→2024–25	Stability	Peri-urban	Large project
1	Chapparadahalli	Bangalore Rural, Karnataka	34.9	4.6→90.7	100%	yes	yes
2	Lachhmanpur	Puruliya, West Bengal	77.4	7.1→164.9	100%	no	yes
3	Banwari Bara	Gautam Buddha Nagar, Uttar Pradesh	43.0	7.9→144.1	100%	no	yes
4	Doddagollahalli	Bangalore Rural, Karnataka	74.2	11.0→97.8	100%	yes	yes
5	Kerki	Saraikela-Kharsawan, Jharkhand	30.0	13.6→187.4	98.8%	no	no
6	Bounker	Ludhiana, Punjab	61.2	39.9→258.3	99.9%	yes	no
7	Muliaguda	Nabarangapur, Odisha	21.1	0.3→10.0	96.8%	no	yes
8	Kishorpur	Gautam Buddha Nagar, Uttar Pradesh	47.4	15.8→342.2	96.8%	no	yes
9	Samletu	Bilaspur, Himachal Pradesh	17.5	1.6→16.2	99.3%	no	yes
10	Palakuppam	Viluppuram, Tamil Nadu	106.3	25.0→191.5	99.5%	no	yes

What drives the top villages. Two patterns stand out. City spillover: Bangalore Rural alone places 10 villages, 9 of them within 10 km of a major town. Single large projects: several flagged villages match known projects (not yet confirmed against imagery), including Banwari Bara and Kishorpur near the Noida International Airport at Jewar, Sajiyali Roopji in Barmer near the HPCL Rajasthan refinery, and Samletu near AIIMS Bilaspur. Villages without either flag, such as Kerki (#5), are the clearest cases of broad local growth.

Quality checks

Each stage was checked for completeness and consistency; the ranking has not yet been validated against independent data.

Check	Result
Every village kept through the name join and GEE run	648,372 in, 648,372 out, unique IDs

Check	Result
Land-cover classes sum to village area (±1%)	All villages except 13 with missing Esri data, which are excluded
Missing or negative values	None, except 5 slivers under 0.5 ha with no light pixel (set to 0)
National totals plausible	Cropland 154 Mha against India's ~140 Mha net sown area; light rises smoothly every year
Top 100 integrity	100 distinct villages, no missing values
Sensitivity to weights	Median top-100 village survives 67% of 1,000 random weight sets

Still to do: before/after imagery checks of the top villages, and comparison of district scores with NFHS-4 → NFHS-5 household asset change.

Limitations

The score measures physical and luminous change, not income.

- **Proxies, not income.** Farm-led growth (better yields, a second crop) barely shows in lights or buildings and is under-ranked.
- **City spillover and large projects are counted as growth.** 31 of the top 100 are within 10 km of a major town and 23 are driven by one large project; both are flagged, not removed.
- **2011 boundaries.** Census 2011 polygons carry ~0–1 km error, and villages absorbed into cities since 2011 still appear as villages.
- **Night-light resolution.** At ~464 m a village covers about 20 pixels; dim rural lights sit near the noise floor, so small absolute changes can look large.
- **Land-cover flicker.** Esri classes can switch between years; the two-year built-up rule reduces this, but 9 top villages still show unexplained crop loss.
- **Judgement calls.** Weights, the 80th-percentile filter and the 2021 light baseline are choices; 29 of the top 100 depend materially on the weights.

How to run

The whole pipeline reruns from the three SHRUG downloads; only step 02 needs an Earth Engine account.

Setup. Python with `pandas`, `geopandas`, `pyogrio`, `numpy` and `earthengine-api`; a Cloud project registered for Earth Engine; run `earthengine authenticate` once. Unzip the SHRUG village, district and state shapefiles from the SHRUG download page into `data/raw/`.

Run order (from the project root)

1. `python scripts/00_add_admin_names.py`
2. `python scripts/01_prepare_gee_upload.py`
3. `python scripts/02_extract_gee.py --project <gcp-project-id>` (about 2.5 hours, resumable; add `--workers 2` if the project hits its quota)
4. `python scripts/03_score_villages.py` (a few minutes; rerun alone to change weights or thresholds)

The scoring choices (years, weights, agreement percentile, flag thresholds) are constants at the top of `03_score_villages.py`.

Outputs

File	Contents
`data/output/top100_villages.csv`	Final top 100: rank, score, stability, location, signals, flags
`data/output/villages_scored.csv`	All 648,372 polygons with signals, score and exclusion reason
`data/interim/village_gee_stats.csv`	Raw GEE statistics per village

Next steps

The most valuable next step is checking the ranking against independent evidence.

- [ ] Before/after imagery check of the top 30 villages, recording real growth vs artefact
- [ ] Compare district-level scores with NFHS-4 → NFHS-5 household asset change
- [ ] Add a farm-growth signal: Sentinel-2 / MODIS cropping intensity

- [ ] Add Census 2011 population for per-capita measures and a better city-distance proxy
- [ ] Map the top 100 for the presentation